

Day 16 Student Handout

Geometry Summer Credit | Circle Vocabulary and Equations

Name:	Date:	Period:
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Learning Targets

I can identify and use circle vocabulary. I can read and write a circle equation in standard form. I can graph a circle from its center and radius and connect the equation to the distance formula.

Lesson 16A: Circle Structure

Warm-Up

1.	Find the distance between (1, 2) and (7, 10).
2.	Simplify: $\sqrt{49}$.
3.	If a circle has radius 6, find its diameter.

Essential Vocabulary

Circle Parts

A circle is the set of all points in a plane the same distance from a center. A radius joins the center to the circle. A diameter passes through the center with endpoints on the circle. A chord has endpoints on the circle. A tangent touches once; a secant crosses twice.

Problem	Work / notes
A circle has radius 9. Find its diameter.	Relationship: Answer:
A circle has diameter 26. Find its radius.	Relationship: Answer:
Explain why every diameter is a chord but not every chord is a diameter.	Shared feature: Required difference:

Standard Equation of a Circle

Center-Radius Form

A circle with center (h, k) and radius r has equation $(x - h)^2 + (y - k)^2 = r^2$. The signs inside the parentheses are opposite the center coordinates. The number on the right is radius squared, not radius.

4. State the center and radius of $(x - 3)^2 + (y + 2)^2 = 25$.

5. Write the equation of a circle with center $(-4, 1)$ and radius 6.

Lesson 16B: Graphing Circles and Algebra 2 Preview

Graph from Center and Radius

Four Anchor Points

Plot the center first. Move r units right, left, up, and down to plot four anchor points. Sketch a smooth circle through them. Check that each anchor point is exactly r units from the center.

Problem	Work / notes
Graph $(x - 2)^2 + (y + 1)^2 = 16$. List the center, radius, and four anchor points.	Center: Radius: Left/right: Up/down:
Write an equation for a circle centered at $(0, 0)$ passing through $(0, -7)$.	Radius: Equation:
Does point $(6, 2)$ lie on $(x - 2)^2 + (y - 2)^2 = 16$? Show substitution.	Substitute: Conclusion:

Completing the Square Preview

Preview Only

To rewrite $x^2 - 6x + y^2 + 4y = 12$, group x -terms and y -terms. Add 9 to complete $x^2 - 6x$ and add 4 to complete $y^2 + 4y$. Add both values to the right: $(x - 3)^2 + (y + 2)^2 = 25$.